1 Voting Theory

1.1 Plurality Method

Definition 1.1.1 A **preference ballot** is a ballot in which each voter ranks the choices according to preference.

Draw out what a simple preference ballot for one voter might look like. What does the total data look like? Note that we usually combine the preference ballots that look similar.

Definition 1.1.2 A **Preference schedule** is a condensed way of writing out several preference ballots.

Draw an example. Notice that the total number of votes is the sum of the numbers above each column.

How do we determine who wins?
Introduce the plurality method!
How many voters are there? Who has the most 1st place votes?

Example 1.1.3 Let's look the following vote.

	1	3	3	3
1st Choice	A	A	O	H
2nd Choice	O	H	H	A
3rd Choice	H	O	A	O

Example 1.1.4 Now you try it!

	10	8	4	3
1st choice	A	B	C	B
2nd choice	B	C	B	A
3nd choice	C	A	A	C

1. How many voters are there?
2. How many first place votes does B have?
3. Who wins under the plurality method?

Example 1.1.5 Let's look the vote from before.

	1	3	3	3
1st Choice	A	A	O	H
2nd Choice	O	H	H	A
3rd Choice	H	O	A	O

What could go wrong here?

What happens if one of the options is no longer available?
Start to do a pairwise comparison. Notice that when H is compared to A, H seems to be preffered, but did not win. Is that fair? What does it mean to have a fair election?

Definition 1.1.6 A **fairness criterion** is a statement which seems like it should be true during an election.

We have now run into our very first Fairness Criterion!

Definition 1.1.7 **(Condercet Criterion)** If there is a choice which is preffered in every one-to-one comparison to the other choices, that choice should win the election. That choice (if it exists) is called the **Condocet Candidate**.

Example 1.1.8 Your turn again! Which option wins the following vote using the plurality method? Is there a Condorcet Candidate? If there is one, did they win the election?

	44	14	70	22	80
1st Choice	G	G	M	M	B
2nd Choice	M	B	G	B	M
3rd Choice	B	M	B	G	G